Opacity as Commitment: Fragile Financial Markets under Delegated Trading*

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Abstract

We study a model of delegated asset management in which a fund manager cares not only about trading profits but also about her reputation, captured by investors' perceptions of the manager's skill. The fund is intentionally made opaque by the manager, who communicates noisy information about her trading strategy to clients. This opacity generates two self-fulfilling equilibria: a benchmark equilibrium and a "flow-driven" equilibrium. In the benchmark equilibrium, the manager prioritizes fund performance, and her trading intensity is set at the profit-maximizing level. In contrast, in the flow-driven equilibrium, the manager trades more aggressively in order to appear skilled, thereby building reputation among investors and attracting larger fund flows, while the fund exhibits poor performance. Our results imply that (1) fund opacity and reputation concerns can trigger market fragility and increase inequality in managers' compensation; (2) fund managers with different skill levels adopt different trading styles; and (3) skill in active management does not necessarily translate into superior performance. Our model also shows that highly skilled fund managers are more likely to choose opaque fund structures, as opacity makes it difficult to communicate trading strategies to investors and weakens incentives for reputation-driven risk-taking. In this sense, fund opacity serves as a commitment device that allows skilled managers to pursue profit-driven trading strategies rather than engaging in flashy risk-taking aimed at signaling high skill.

JEL classification: G11, G12, G14, G23

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1 Introduction

XXX Cite Prat (2005, AER)/ Gervais and Strobl (2019, RFS) XXX Financial firms compete for skilled employees. As the boundaries between investment banking, commercial banking, and asset management blur, this competition intensifies. Following the 2007-2009 crisis, pressures to reduce compensation and the resulting challenges in retaining talent have brought poaching practices into focus.[1] Poaching, or more generally labor mobility, raises concerns among financial specialists about their reputation within the industry. How do these concerns influence their trading strategies and skill acquisition? How do they interact with asset prices and market liquidity?

To address these questions, we study a model inspired by Kyle (1985), where traders, privately informed about their skill, trade risky assets anonymously with market makers. We define skill as the trader-specific ability to uncover more information about asset payoffs than the market. The more skilled a trader is, the larger the potential mispricing she can identify, and thus, the greater the trading profits she can potentially earn. Each trader selects an asset based on her skill and strategically submits a market order to the market maker for that asset, who sets the price after observing the aggregate order flow. A recruiter attempts to infer each trader's skill from the observed price and payoff of the asset chosen by the trader. The trader values not only trading profits but also her reputation, i.e., the recruiter's estimate of her skill. In a simple one-period model presented in Section 2, each trader derives an exogenous benefit from her reputation, which we model as an increasing quadratic function of reputation. In Section **??**, we extend the model to a two-period setting, providing a microfoundation for the reputation benefit. In this extension, each trader randomly encounters a recruiter who compensates

[1] According to *The Wall Street Journal*, poaching is a common practice in the sophisticated capital markets industry, occurring in both bull and bear markets ("Greed is Good," February 2009), and employees at all levels are being poached ("Employee Poaching Returns to Wall Street," May 2010). *The New York Times* ("A Mad Scramble for Young Bankers," July 2014) and *Bloomberg* ("Saturdays Off Lose to Cash as Buyout Firms Poach Junior Bankers," March 2014) highlight the buy-side's (private equity, hedge funds) recruitment of young talent from investment banking amid a changing financial industry.

her for her skill based on his estimate. In equilibrium, the quadratic form assumed in Section 2 arises endogenously.

The model yields two stable, self-fulfilling equilibria, one of which is similar to that of Kyle (1985), where reputation is irrelevant. In the other, the "reputation-driven" equilibrium, traders trade more aggressively to appear skilled. This strategy increases trading volume, price volatility, and market liquidity but undermines trading profits by revealing too much private information to the market maker. Intuitively, each trader can boost her reputation by placing a large order, either buy or sell, because such an action makes her appear like a skilled trader who has identified an asset with significant mispricing. In equilibrium, the recruiter accurately anticipates the trader's strategy, so the recruiter's learning is not manipulated; nevertheless, the trader may trade aggressively depending on the market's belief. To illustrate the mechanism, suppose the recruiter believes that the trader trades more aggressively than the profit-maximizing level. If she trades less aggressively to increase profits, the recruiter (incorrectly) infers that she is less skilled. To avoid the reduction in her reputation benefit, she conforms to the recruiter's belief and indeed trades very aggressively.

The self-fulfilling property of our equilibria implies that a mere shift in investor beliefs, unaccompanied by changes in fundamentals, can trigger a transition from the standard Kyle-like equilibrium to the reputation-driven one, where price volatility is high and trading performance is relatively poor. We interpret this as *fragility* in financial markets.[2] Key to our multiple equilibria and fragility results are the trader's conflicting motives for revealing skill: on one hand, she wishes to hide her skill from the market maker to earn trading profits, but on the other hand, she wants to show it off to the recruiter to obtain the reputation benefit. If the trader's concern for reputation is very weak (very strong), the first (second) motive dominates, and the Kyle-like (reputation-driven) equilibrium holds uniquely. However, if the concern is moderately strong, either equilibrium may arise depending on the market's

[2] As in Greenwood and Thesmar (2011), we define a market as fragile if it is susceptible to non-fundamental demand shifts, such as those caused by changes in investor beliefs.

self-fulfilling beliefs. Therefore, the model predicts that fragility may occur when poaching is (moderately) active and there is high demand for financial skill, as these situations increase the likelihood of future recruitment and lead to heightened reputation concerns.

Our results shed light on how traders' underlying skills relate to their trading styles and contribute to financial fragility. The model predicts that high-skilled traders adopt low-intensity, profit-driven strategies, while those with lower skills engage in aggressive, risk-seeking behavior to gain reputation. Traders with intermediate skill levels, facing multiple equilibria, may switch between profit-seeking and reputation-driven strategies based on market beliefs, thereby contributing to market fragility. This behavior aligns with real-world observations during financial crises, such as momentum crashes and sudden reversals, which occur even without fundamental shocks.

The model has implications for compensation in the finance industry. In the reputation-driven equilibrium, traders' wages exhibit greater inequality among high- and low-skilled traders compared to those in the Kyle-like equilibrium, as they trade aggressively and reveal their skill more than they do in the Kyle-like equilibrium. This allows the recruiter to better assess each trader's skill and, therefore, to offer wages that more accurately reflect their abilities. This result is consistent with the findings by Philippon and Reshef (2012) and Célérier and Vallée (2019), who document increasing returns to talent and their significant contribution to income inequality in the finance industry.

The aggressive trade in the reputation-driven equilibrium also causes traders to reveal excessive information to the market maker, resulting in poorer trading performance than in the Kyle-like equilibrium. Consequently, high-skilled traders in the reputation-driven equilibrium may underperform low-skilled traders following the Kyle-like equilibrium. This result offers a novel explanation for the puzzling relationship (or the lack thereof) between trader skill and persistent performance, as documented and analyzed in the literature (e.g., Busse, Goyal, and Wahal, 2010; Berk and Green, 2004).

In Section **??**, we study traders' endogenous skill acquisition. We show that, due to the multiplicity of equilibria, each trader may intentionally acquire a "middling" level of skill, which is too high or too low compared to the level she would acquire if she knew which type of equilibrium would prevail. Ex-post, in the Kyle-like equilibrium, where profits are prioritized over reputation, the acquired skill level is insufficient. Conversely, in the reputation-driven equilibrium, where reputation outweighs profit motives, the skill level is excessive. Ex-ante, without knowing which equilibrium will emerge, the trader optimally chooses a middling skill level to prepare for both scenarios. This finding aligns with real-world observations that some finance professionals tend to pursue broader business qualifications rather than specializing in areas like advanced computer science or pursuing higher academic degrees such as a PhD.

Our paper is related to the literature on fragility in financial markets. Cespa and Vives (2015) is most closely related, as they also find multiple equilibria in an asymmetric information trading model. The key ingredients of their model are short-term investor horizons and persistent liquidity trading, which generate strategic complementarities. In Froot, Scharfstein, and Stein (1992), strategic complementarities among short-term traders are also important for the emergence of multiple equilibria. Unlike these studies, our model does not feature short horizons; instead, we obtain fragility from traders' forward-looking concerns about future recruitment opportunities. In Cespa and Foucault (2014), learning from other assets amplifies shocks and causes liquidity crashes. Although the mechanism is different, our model can also result in liquidity crashes when the economy switches between equilibria due to non-fundamental shocks.

Our paper also connects to the literature on trading frenzies, such as Veldkamp (2006) and Goldstein, Ozdenoren, and Yuan (2013), in which strategic complementarities play a key role. In Veldkamp (2006) frenzies originate from complementarities in information markets, while in Goldstein, Ozdenoren, and Yuan (2013) strategic complementaries, triggered by a feedback effect from financial markets to a firm's real investment, lead to coordinated aggressive trading. By contrast, in our reputation-driven equilibrium, each trader trades aggressively out of fear of being perceived as less skilled by the recruiter who believes that she would trade aggressively.

Lastly, our work contributes to the broader understanding of the role of asymmetric information about skills in financial markets. DiMaggio (2015), Malliaris and Yan (2021), and Bijlsma, Boone, and Zwart (2018) derive implications for the risk-taking behavior of agents when ability is private information but is reflected in performance. We also link trader skill to private information and asset markets; however, unlike these papers, our focus is on deriving pricing implications. Guerrieri and Kondor (2012) show that career concerns amplify price volatility, similar to our findings. In Dasgupta (2006), career concerns lead to churning (trading without private information), and in Dasgupta and Prat (2008), they lead to herding (ignoring private information and trading like everyone else). These three papers are closely related to ours in that they study career concerns and endogenous asset prices, but they do not analyze fragility, which is the main focus of our paper.

The rest of the paper proceeds as follows. Section 2 presents a baseline model with exogenous reputation concerns, and Section 3 studies the equilibrium. Section **??** endogenizes reputation concerns by introducing a two-period model and presents the implications of our results. Section **??** analyzes traders' skill acquisition. The Online Appendix contains all proofs for the theoretical results.

2 Model

This section presents a one-period trading model à la Kyle (1985) with delegated asset management.

2.1 Setting

The economy consists of a *fund manager*, a competitive *market maker*, a *noise trader*, and a continuum of *investors*. All players are risk neutral. The manager leverages her skill (defined later) to identify a single risky asset with potential mispricing, raises capital from investors, and allocates it between the risky asset and a risk-free asset with a zero interest rate.[3]

2.1.1 Skill

The manager can uncover more information about the asset's payoff than the market maker, and we interpret it as her *skill*. At the beginning of the period, the manager draws private information s from a normal distribution with mean zero and variance ω_s. Leveraging this information, she identifies an asset with payoff $\delta = \bar{\delta} + s$, where $\bar{\delta}$ is a publicly know mean payoff. Since the market maker does not observe s, her prior expectation of the asset's payoff is $\mathrm{E}[\delta] = \bar{\delta} + \mathrm{E}[s] = \bar{\delta}$, which is biased by s from the informed manager's perspective. As the market maker would set a price $p = \mathrm{E}[\delta] = \bar{\delta}$ in the absence of traders, s can be viewed as the asset's "potential mispricing."[4] Indeed, as shown later, the manager profits by buying the asset if $s > 0$ (underpriced) and selling it if $s < 0$ (overpriced); the larger the $|s|$, the larger the profit. In light of this, we measure the manager's *realized* asset-selection skill by $|s|$, while the variance ω_s serves as an ex-ante measure of the average skill.[5,6]

2.1.2 Capital Allocation and Trading

We model the manager's trading decision and investors' capital allocation as a simultaneous game with rational expectations, as illustrated in Figure 1.

[3] We abstract away from information acquisition by individual investors and assume that they invest through a manager due to the lack of informational advantages. This assumption would be supported by, for example, information- or skill-acquisition costs, where only the fund manager acquires the skill at lower costs due to economies of scale (e.g., Gârleanu and Pedersen, 2018).

[4] s is not the *actual* mispricing because the market maker will take into consideration the presence of the informed manager; thus, p will deviate from $\bar{\delta}$ in equilibrium.

[5] This formalization of skill aligns with findings in the empirical studies that attribute fund managers' performance to stock-picking skills (e.g., Wermers, 2000; Kosowski, Timmermann, Wermers, and White, 2006).

[6] The definition of the average skill follows from $\mathrm{E}[|s|] = \omega_s \sqrt{\frac{2}{\pi_0}}$, where π_0 denotes the circular constant.

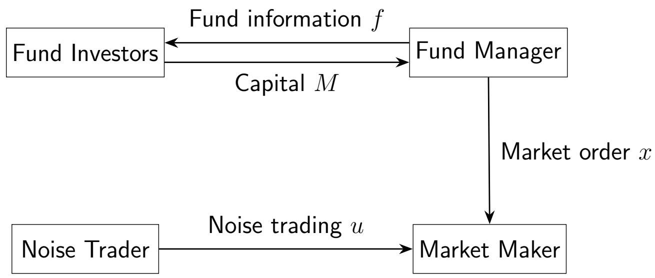


Figure 1: Capital Allocation and Investment

Manager's tradign decision. Upon discovering the mispricing in the asset s, the fund manager chooses the portfolio allocation between the risky and the safe asset. This action is represented by a market order for $x \in \mathbb{R}$ units of the risky asset, submitted to the market maker in the financial market. The manager decides on x by incorporating the reactions of the financial market (i.e., the asset price) and the capital inflow from her investors, defined below.

Investors' capital allocation. Each investor is endowed with one unit of wealth and decides whether to invest into the fund. Investor i's decision is denoted by $m_i = 1$ (invest) and $m_i = 0$ (not invest), and the total fund flow or the asset under management (AUM) of the fund is defined by $M \equiv \int m_i di$.[7] The fund's fee structure is proportional to the total fund proceeds: the manager takes a $\phi \in (0, 1)$ fraction of the proceeds, while the remaining $1 - \phi$ fraction is distributed to the investors in proportion to their contributions to the fund, i.e., m_i/M.[8]

Financial market. In the financial market, the market maker sets the asset price, p, based on the incoming order flow, $x + u$, where u represents a random market order from the noise trader and follows a normal distribution with mean zero and variance ω_u. Due to competition, the market maker sets a

[7] Modeling the investor's decision as a continuous choice rather than a discrete one yields the same results, as the objective function is linear in m_i (see equation [2.4]).

[8] Introducing a fixed fee does not change our qualitative results, as long as the performance-based fee also exists, $0 < \phi < 1$.

semi-strong efficient price conditional on $x + u$:[9]

$$p = \mathrm{E}[\delta | x + u]. \tag{2.1}$$

2.1.3 Fund Opacity

At the fund-raising stage, the manager disseminates a noisy signal about her trading strategy to investors, defined as $f = x + e$, where e represents a normally distributed noise term with mean zero and variance ω_e. This signal is referred to as the *fund information*, and the noise variance ω_e is referred to as *fund opacity*. This signal is conveyed through private interactions between the manager and her investors and can be interpreted as the fund manager's practice of communicating her strategy, such as a high-conviction or concentrated risky position, to investors in order to attract capital.[10] Since the manager chooses her trading strategy x based on the private signal s, the fund information f is informative about the manager's skill and the fund's return. Thus, fund investors use f to decide on their capital allocations. In what follows, we first consider the baseline model with a perfectly transparent fund ($\omega_s = 0$), while Section 4 deals with general opacity ($\omega_s > 0$). Section 5 considers the manager's strategic choice of fund opacity.

2.1.4 Misreporting and Verification.

The fund's profit from the risky investment is represented by $\pi \equiv (\delta - p)x$, so that the total fund proceeds, including those from the safe asset, are

$$y = \pi + M. \tag{2.2}$$

The fund manager, however, has an incentive to make a false report about y to her investors. Namely, by reporting $\tilde{y}$, she pays out $(1 - \phi)\tilde{y}$ to investors

[9]Assume that, on the equilibrium path, one competitive market maker trades the asset. As in Kyle (1985), competitively many market makers exist off the equilibrium path, ensuring a break-even price in the equilibrium.

[10]Such activities are consistent with the empirical evidence and theoretical rationale in Anton, Cohen, and Polk (2021) and Van Nieuwerburgh and Veldkamp (2010).

and captures $y - (1-\phi)\tilde{y}$ on her own, suggesting that reporting $\tilde{y} = 0$ would be optimal.[11] To prevent this intentional misreporting, we assume that each investor pays the *verification cost* C if she invests in the fund. This cost gives investors the ability to verify (i.e., back out) the manager's investment strategy x from the fund information in the end of the investment period, when they also observes the market data on the asset's price p and payoff δ. Otherwise, the manager can intentionally misreport $\tilde{y} = 0$, as there are levels of noise in the fund information (e) and noise trading (u) that are consistent with the report, i.e., the manager claims that the investment was $\tilde{x} \equiv -M/(\delta - p)$, while realized noise terms were $\tilde{e} = f - \tilde{x}$ and $\tilde{u} = (x+u) - \tilde{x}$, making f and p consistent with $\tilde{y} = 0$. For tractability, we assume that the verification cost for each investor is specified as $C \equiv \frac{c(\omega_e)}{M}$ with $c'(\cdot) > 0$ and $c(0) = 0$. Namely, the verification cost is increasing in the degree of fund opacity, while costless verification is possible if the fund is perfectly transparent. This assumption follows from the intuition that verification requires more intensive analyses of fund information f when the fund is more opaque. Moreover, the verification cost is shared among fund investors with measure M. This is realistic, as the verification in reality often involves collective auditing and, once information about x is uncovered, it is a common good shared by all investors.[12]

2.1.5 Maximization Problems and Equilibrium

Given the realized skill s, the fund manager maximizes her expected fee income by controlling her investment strategy x:

$$\max_x \mathrm{E}\left[\phi y | s\right]. \tag{2.3}$$

[11] Technically, the manager earns an infinite payment by reporting $\tilde{y} = -\infty$, while we assume that investors must receive non-negative payoffs from the fund (i.e., limited liability), thereby imposing $\tilde{y} \geq 0$.

[12] The qualitative results ramins unchanged if we assume that a part of the verification cost cannot be shared by investors.

Conversely, conditional on the fund information f, investor i maximizes her expected return from delegating capital management:

$$\max_{m_i \in \{0,1\}} \mathrm{E}\left[\frac{m_i}{M}(1-\phi)y - m_i - Cm_i | f\right]. \tag{2.4}$$

Note that the first term represents the expected payment from the fund, the second term is the cost of capital, and the last term is the verification cost, which becomes relevant only when she makes investment ($m_i = 1$).

Definition 1. *The equilibrium is defined by the market maker's pricing strategy p, the fund manager's trading strategy x, and investors' capital allocations $\{m_i\}_{i=1}^{\infty}$, such that, (i) p satisfies (2.1) given $x+u$, (ii) x solves (2.3) given s, p, and $\{m_i\}_{i=1}^{\infty}$, and (iii) for all i, m_i solves (2.4) given p, f, and $\{m_j\}_{j\neq i}$.*

3 Equilibrium

In this section, we search for a linear equilibrium under transparency ($\omega_e = 0$), in which the manager's market order x is linear in her skill s, and the market maker's pricing strategy p is also linear in the order flow $x + u$. Proofs for the analytical results are provided in the Appendix.

3.1 Trading Strategy

Conjecture and later verify that the manager's equilibrium trading strategy is

$$x(s) = \beta s, \tag{3.1}$$

where $\beta > 0$ is an endogenous constant determined in the equilibrium. Conjecture (3.1) states that the manager buys (sells) the asset if her s is positive (negative), where the order size is proportional to her skill $|s|$; the more skilled the manager, the more aggressively she trades.[13] The scale factor β is referred

[13] This conjecture follows from the original Kyle model where an informed trader trades based on the difference between her belief and that of the market maker, i.e., $\mathrm{E}[\delta|s] - \mathrm{E}[\delta] = s$.

to as the *trading intensity*, representing how intensively the manager exploits her informational advantage over the market maker.

3.2 Execution Price

The market maker decides on the price of the asset by extracting information about the asset's payoff from the aggregate order flow $x+u$. Anticipating the trading strategy (3.1), the price in equation (2.1) can be rewritten as

$$p = \bar{\delta} + \lambda(x+u), \tag{3.2}$$

where

$$\lambda \equiv \frac{\beta\omega_s}{\beta^2\omega_s + \omega_u} \tag{3.3}$$

is referred to as "Kyle's lambda," a measure of the price impact of order flow.[14] Following the literature, we define a market to be *liquid* if λ is small.

3.3 Investors' Capital Allocation

By incorporating the fund proceeds $y = \pi + M$, investor i's optimization problem in (2.4) is

$$\max_{m_i \in \{0,1\}} \frac{m_i}{M}(1-\phi)\mathrm{E}\left[\pi|f\right] - (\phi + C)\, m_i. \tag{3.4}$$

Since each investor is infinitesimal, she does not incorporate the impact of her capital allocation (m_i) on the total fund flow (M) in solving (3.4). Therefore, the investor's objective function is linear in m_i, while the marginal benefit of fund participation is diminishing in M, as the fund's proceeds are allocated to investors on a pro-rata basis. This structure pins down the optimal capital

[14] From (2.1), we have $p = \mathrm{E}[\delta|x+u] = \mathrm{E}[\mathrm{E}[\delta|x+u,s]|x+u] = \mathrm{E}[\bar{\delta}+s|x+u]$. Since the market maker believes that the manager follows (3.1), s and $x+u$ are both normally distributed from the market maker's perspective. Thus, $p = \bar{\delta} + \mathrm{E}[s] + \frac{\mathrm{Cov}[s,x+u]}{\mathrm{Var}[x+u]}(x+u-\mathrm{E}[x+u]) = \bar{\delta} + \frac{\mathrm{Cov}[s,\beta s+u]}{\mathrm{Var}[\beta s+u]}(x+u-\mathrm{E}[\beta s+u]) = \bar{\delta} + \frac{\beta\omega_s}{\beta^2\omega_s+\omega_u}(x+u)$.

allocation by investor i as follows:

$$m_i = \begin{cases} 1 & \text{if } \frac{1}{M}(1-\phi)\mathrm{E}[\pi|f] > \phi + C, \\ \in \{0,1\} & \text{if } \frac{1}{M}(1-\phi)\mathrm{E}[\pi|f] = \phi + C, \\ 0 & \text{otherwise,} \end{cases} \tag{3.5}$$

where the middle case indicates that the investor is indifferent between investing and not. We focus on the equilibrium where the total fund flow M satisfies this indifference (or the free-entry) condition, meaning that all investors break even. This value of M is stable: as long as M is smaller than this critical value, the additional investor would earn strictly positive returns by delegating capital management, meaning that the fund attracts larger fund flows and M keeps growing, while once it exceeds the critical value, the delegated asset management leads to negative returns, triggering capital withdrawal and a decline in M. In this equilibrium, the total fund flow is determined by

$$M = \theta\mathrm{E}[\pi|f] - \frac{c(\omega_e)}{\phi}, \tag{3.6}$$

where $\theta \equiv (1-\phi)/\phi$. It is linearly increasing in the expected trading profit conditional on the fund information, $\mathrm{E}[\pi|f]$, while the verification cost discourages investors from investing into the fund. The fund flow positively depends on $\mathrm{E}[\pi|f]$ due to the performance-based fee structure, which makes the per-unit return from delegated asset management increasing in the trading profit. Also, equation (3.6) specifies the coefficient of fund flow by θ, which we refer to as the *flow-performance sensitivity.* It governs the reaction of the fund flow to the expected fund performance and is increasing in the fee rate (ϕ), as high ϕ raises the cost of delegation and discourages investment.

To compute the expected fund profits, each investor seeks to back out the realized skill level of the manager $|s|$ from fund information $f = x + e$. When the fund information is perfectly transparent (i.e., $\omega_e = 0$), investors rationally use the trading strategy in (3.1) and infer $s = x/\beta$, leading to[15]

$$\mathrm{E}\left[\pi|f\right] = (1-\lambda\beta)\beta \mathrm{E}[s^2|s = x/\beta] = \left(\frac{1}{\beta} - \lambda\right) x^2. \tag{3.7}$$

The first equality in (3.7) suggests that the fund return, $\pi = (\delta - p)x$, becomes a quadratic function of s, as both the manager's trading quantity x and the profit margin, $\delta - p$, are proportional to the skill level in expectation. Furtheremore, the transparent fund information indicates that $s = x/\beta$, thereby shaping the conditional expected profit into quadratic in x, as the second equality in (3.7) attests.

3.4 Manager's Optimization

By incorporating the total fund flow ([3.6] and [3.7]), the manager's problem in (2.3) is re-written as

$$\max_{x} \phi \left(sx - \lambda x^2 + \theta \left(\frac{1}{\beta} - \lambda\right) x^2 \right). \tag{3.8}$$

The first two terms represent the fee income from the trading profit and are essentially identical to the trading profit in the original Kyle (1985) model. The third term arises due to the fund inflow and, as equation (3.7) demonstrates, is determined by the quadratic size of risky investments. The optimal trading strategy solves (3.8), leading to

$$x = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1-\lambda\beta)} s. \tag{3.9}$$

It is consistent with our guess in (3.1) when:

$$\beta = \frac{1}{2} \frac{\beta}{\lambda\beta - \theta(1-\lambda\beta)}. \tag{3.10}$$

[15] Given the linear pricing rule in (3.2), it holds that $\pi = (\delta - p)x = (s - \lambda\beta x)x$. Adopting trading strategy $x = \beta s$ leads to equation (3.7).

By solving λ and β in equations (3.3) and (3.10), the trading strategy and the asset price are uniquely determined:

Lemma 3.1. *When the fund information is transparent ($\omega_e = 0$), the manager's trading strategy and the asset price are specified by* (3.1) *and* (3.2) *with*

$$\beta = \sqrt{(1+2\theta)\frac{\omega_u}{\omega_s}}, \tag{3.11}$$

and

$$\lambda = \frac{1}{2(1+\theta)}\sqrt{(1+2\theta)\frac{\omega_s}{\omega_u}}. \tag{3.12}$$

In the absence of delegated asset management ($\theta \to 0$), the equilibrium is identical to the original Kyle (1985) model, where β is determined to maximize the expected trading profits. On one hand, the manager seeks to exploit her informational advantage by trading more intensively. On the other hand, it reveals information to the market maker and raises the price impact, cutting into the trading profit. The equilibrium β balances these competing effects.

Delegated asset management ($\theta > 0$) brings about the additional risk-taking incentive, because inflating x benefits the manager by making her appear highly skilled and inducing a larger fund flow (see equations [3.6] and [3.7]). We refer to this incentive as the "showing-off" motive. Since θ governs the flow-performance sensitivity, this motivation becomes stronger when θ is large, shaping equilibrium β into an increasing function of θ.

Equation (3.12) shows that the price impact λ decreases with θ. This is because the information is exaggerated in that the order size is too large relative to the true $|s|$ due to the showing-off motive. Thus, the market maker attempts to partially undo the manager's information revelation by incorporating less of it into the price, resulting in a weaker price impact.[16]

Our benchmark result in Lemma 3.1 isolates the fundamental mechanism through which delegated asset management enhances the manager's risk taking behavior. The tendency of fund managers to show off their skill to attract

[16] λ represents the price's reaction to the order flow $x + u$, rather than its reaction to the manager's private information s. The price impounds s with coefficient $\beta\lambda$, which is monotonically increasing in θ.

capital is well discussed in the literature (e.g., Metrick and Yasuda, 2010; Chung, Sensoy, Stern, and Weisbach, 2012; Barber and Yasuda, 2017), raising regulatory concerns that fund managers have incentives to exaggerate the performance or quality of their funds. In our model, however, a skilled manager may also have incentives to conceal her skill from the market maker, as emphasized by Kyle (1985). The next section explores how the fundamental tension between the showing off and the hiding motives shapes the equilibrium when the fund information becomes harder to interpret for investors due to opacity.

4 Opaque Funds

This section introduces the fund opacity by assuming $\omega_e > 0$ and demonstrates that belief-driven mltiple equilibria show up regarding the trading strategy of the fund. We discuss economic implications drawn from the equilibrium analyses in Subsection 4.4.

4.1 Delegation under Opacity

The analysis of an opaque fund largely follows the benchmark case in Sections 3 and 2. In particular, investors' optimization and the structure of the resulting fund flows are identical to M in equation (3.6). However, due to the noisy revelation of the fund information, investors' estimation about the manager's skill s and thus the expectation about the fund return must be modified as follows:

Lemma 4.1. *Upon observing the fund information $f = x+e$, investors update their expectation about the trading profit as*

$$\mathrm{E}\left[\pi|f\right] = (1-\lambda\beta)\beta\left(\hat{s}^2 + \hat{\omega}_s\right), \tag{4.1}$$

where the posterior expectation of the (signed) realized skill is

$$\hat{s} = \mathrm{E}[s|f] = \xi(x+e), \tag{4.2}$$

with

$$\xi = \frac{\beta\omega_s}{\beta^2\omega_s + \omega_e}, \tag{4.3}$$

and the posterior of the skill variance is

$$\hat{\omega}_s = \mathrm{Var}[s|f] = \frac{\omega_s\omega_e}{\beta^2\omega_s + \omega_e}. \tag{4.4}$$

Equation (4.2) describes how investors update their belief about the manager's skill using $f = x + e$, where $\hat{s}$ is referred to as the *reputation* in the following discussion. According to equation (4.2), ξ measures how much investors rely on f in updating their belief and governs the sensitivity of the reputation to the fund's trading strategy x. Holding β constant, ξ is large when investors' prior is unreliable (ω_s is large) and the fund information is transparent (ω_e is small). More importantly, ξ itself also depends on the trading strategy β, as a more intensive trading strategy makes f more informative about s.

4.2 Manager's Optimization

Following the same procedures as the transparent benchmark, the manager's optimization problem in (3.8) is re-written as follows:

$$\max_x \; sx - \lambda x^2 + \theta(1 - \lambda\beta)\beta\xi^2(x^2 + \omega_e), \tag{4.5}$$

where we omit the terms unrelated to x. As in the fully transparent benchmark in Section 3, the first two terms of equation (4.5) represent the expected trading profit, and the third term corresponds to her fee income from fund flow M. Similar to the benchmark case, the total fund flow increases with the manager's risk taking, measured by x^2, because she can potentially inflate her reputation $|\hat{s}|$ by placing a large order (either buy or sell). Contrary to the transparent case, however, the reaction of the fund flow to the trading strategy is now influenced not only by the flow-performance sensitivity θ but also by the investors' updating coefficient ξ. This is because investors cannot perfectly observe the manager's trading strategy when the fund is opaque. Importantly, investors' estimate of the manager's skill and, therefore, the reaction of the fund flow to the risky investment depend on the risky trading strategy itself: the more intensive the manager invests, the more reliable the fund information becomes for investors. This structure is captured by the dependence of ξ on β in equation (4.3) and leads to the strategic complementarity between the manager's risk taking and her reputation held by investors.

The unique solution to the manager's optimization problem in (4.5) (given β) is represented by

$$x = F(\beta)s, \tag{4.6}$$

where

$$F(\beta) \equiv \frac{1}{2\left(\lambda - \theta(1-\lambda\beta)\beta\xi^2\right)}. \tag{4.7}$$

$F(\cdot)$ is a nonlinea function of β due to the dependence of λ and ξ on β.

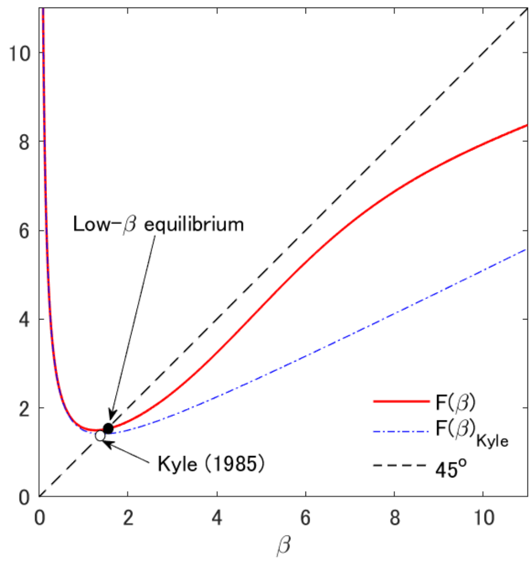
4.3 Equilibrium under Opacity

The manager optimally chooses (4.6) given that the market maker and investors believe that she follows strategy (3.1). In equilibrium, their beliefs must be correct, meaning that (3.1) and (4.6) must be consistent with each other. This belief consistency condition requires that

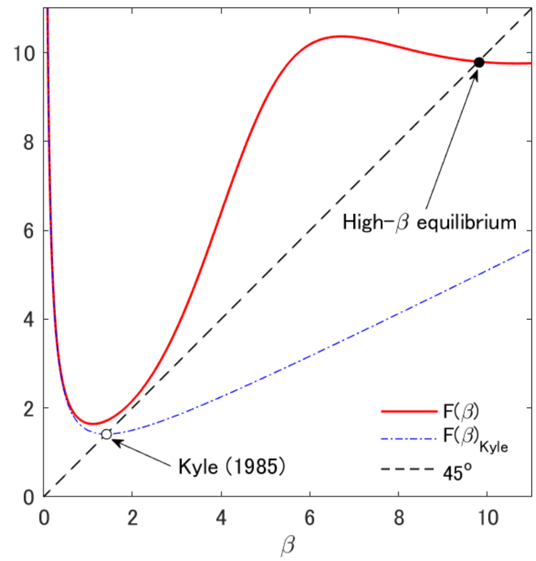
$$\beta = F(\beta). \tag{4.8}$$

That is, the equilibrium levels of β are determined at the fixed points of $F(\cdot)$. Having identified β, the equilibrium levels of x, p, and M are determined by equations (3.1), (3.2) and (3.6), respectively. In what follows, we first illustrate the determination of equilibrium β through a numerical example, followed by an analytical proposition that formally establishes the equilibria.

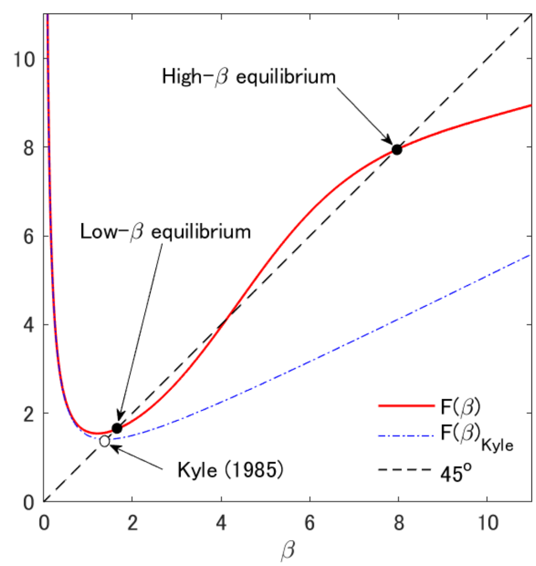
Figure 2 illustrates the determination of β for three different levels of fund opacity ω_e. The intersections of $F(\beta)$ (the solid line) and the 45-degree line yield the equilibrium levels of β. We also plot $F(\beta)_{\text{Kyle}}$ with $\theta = 0$ in the dash-dotted line, representing the original Kyle (1985) model, where fund delegation



(a): Large ω_e ($\omega_e = 48$) $\Rightarrow$ unique low-β equilibrium.



(b): Small ω_e ($\omega_e = 28$) $\Rightarrow$ unique high-β equilibrium.



(c): Intermediate ω_e ($\omega_e = 38$) $\Rightarrow$ multiple high-β & low-β equilibria.

Figure 2: Determination of β

Note: The figure plots $F(\beta)$ for different values of θ. The equilibrium values of β are determined at the fixed points of $F(\cdot)$. The parameter values used in the figures are $\omega_s = 1$, $\omega_u = 2$, and $\phi = 0.02$.

is irrelevant.

Panel (a) of Figure 2 shows that if ω_e is large, there is a unique "low-β equilibrium," in which β is close to the trading intensity without delegated asset management. Panel (b) shows that a large ω_e leads to a unique "high-β equilibrium," in which β is much larger than that in the original Kyle (1985) model. Panel (c) shows that multiple equilibria arise for an intermediate level of ω_e. Specifically, the high-β and the low-β equilibria are stable, as $F(\beta)$ crosses the 45-degree line from above, whereas the one with intermediate β is unstable, as $F(\beta)$ crosses the line from below.[17] In what follows, we focus only on the stable equilibria.

Impact of fund opacity. To see the intuition behind the different equilibrium patterns presented in Figure 2, note that the fund manager faces two conflicting motives when trading. On the one hand, she seeks to avoid trading too aggressively on her private information, as doing so would move the price excessively and reduce her trading profits. As in Kyle (1985), this motive leads the manager to decrease β. On the other hand, she wishes to trade intensively to signal her skill to her investors, aiming to establish a strong reputation and attract large fund flows. This motive encourages her to increase β. In summary, she faces a tradeoff between "hiding" her skill from the market maker and "showing it off" to her investors. The fund opacity ω_e governs this tradeoff by influencing the sensitivity of the manager's reputation held by investors to her investment behavior. In particular, Lemma 4.1 demonstrates that the fund flow M becomes less sensitive to x under opacity (i.e., ξ is small) compared to the fully transparent case, as it becomes difficult for investors to estimate the skill based on the fund information f. Therefore, fund opacity diminishes the strategic complementarity between the manager's risk taking and the reputation, thereby weakening the showing-off motive.

For an opaque fund with a large ω_e (panel [a]), the hiding motive domi-

[17] We define an equilibrium to be stable (unstable) if the corresponding β is a stable (unstable) fixed point of $F(\cdot)$, that is, $|F'(\beta)| < 1$ (> 1). A stable equilibrium is robust to a small perturbation to the agents' belief about β, meaning that the economy converges back to the original equilibrium through the best-response dynamics of the agents' beliefs.

nates the showing-off motive, leading to a unique low-β equilibrium. In this equilibrium, investors believe that the manager's trading intensity is relatively low and the fund information is not very informative, making the reputation and the fund flow insensitive to x. Although the manager could inflate her reputation by placing an order larger than what her investors anticipate, she refrains from doing so, because it would cause a large price impact and reduce the fund's investment profits, while fund flows and her fee income would not increase much due to small ξ. Consequently, the equilibrium β is close to the trading intensity without delegated asset management, meaning that it is a "Kyle-like equilibrium."

For a transparent fund with a small ω_e (panel [b]), the showing-off motive prevails, resulting in a unique high-β equilibrium. In this equilibrium, investors believe that the manager is trading aggressively. The reputation and fund flows grow sensitive to the manager's trading strategy through the fund information. Under such beliefs, the manager is actually willing to trade aggressively: although it moves the price substantially and undermines investment profits, it helps her build a strong reputation and generates a large fee income from significant capital inflows. Importantly, investors' learning is not manipulated, as their belief about the manager's trading strategy is correct. Nonetheless, the manager places a large order because doing otherwise would induce her investors to incorrectly estimate that she is less skilled, thereby deteriorating her reputation. She trades aggressively solely to conform to her investors' belief and to attract large fund flows. To emphasize this mechanism, we call this equilibrium the "flow-driven equilibrium."

For intermediate levels of ω_e (panel [c]), the strategic complementarity between the reputation and the manager's risk taking is neither weak nor strong, and there is no clear dominance between the two motives. Consequently, both the low-β and the high-β equilibria are supported under intermediate fund opacity. Each equilibrium is associated with a self-fulfilling belief of the market maker and investors.[18]

[18] For simplicity, we assume that agents do not agree to disagree, so that they coordinate their belief when parameter values are such that multiple equilibria exist.

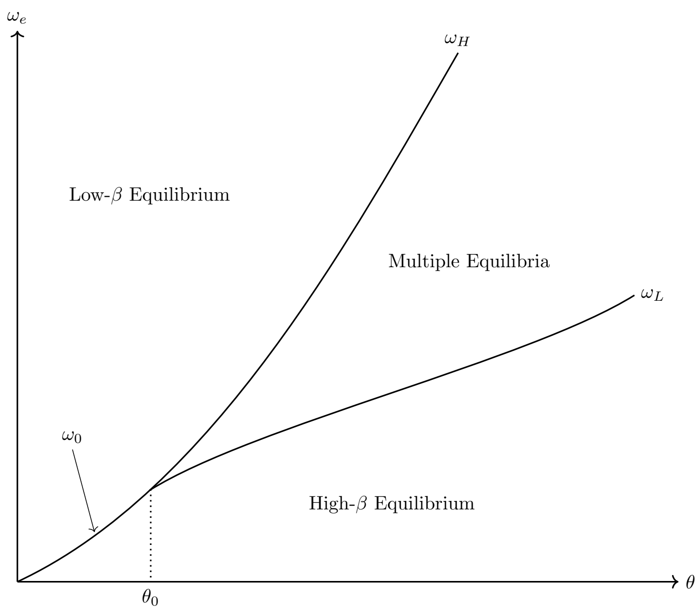


Figure 3: Equilibrium Characterization

Note: This figure illustrates the equilibrium states using the flow-performance sensitivity (θ) and fund opacity (ω_e). When $\theta \leq \theta_0$, the cutoffs of opacity converge to $\omega_H = \omega_L = \omega_0$.

The following proposition summarizes the above discussion and provides the analytical characterization of the equilibria.

Proposition 4.1 (Equilibrium characterization)**.** *There exists a critical value of the flow-performance sensitivity, denoted as θ_0.*

(i) If $\theta \leq \theta_0$, the equilibrium is unique and characterized by a threshold of fund opacity ω_0, such that, $\omega_e \leq \omega_0$ leads to the high-β equilibrium and $\omega_e \geq \omega_0$ leads to the low-β equilibrium.

(ii) If $\theta > \theta_0$, there are two thresholds of fund opacity, ω_L and $\omega_H (> \omega_L)$. When $\omega_e \leq \omega_L$, the high-β equilibrium is unique; when $\omega_e \geq \omega_H$, the low-β equilibrium is unique; and when $\omega_e \in (\omega_L, \omega_H)$, the high- and low-$\beta$ equilibria coexist.

Figure 3 illustrates the result in Proposition 4.1. It indicates that the flow-performance senitivity θ is also critical in determining the equilibrium.

The intuition follows from Lemma 3.1 with transparent fund information: it influences the reaction of the fund flow to changes in the manager's trading strategy, thereby affecting the showing-off motive of the manager. As high levels of θ strengthen this motive, we obtain the following corollary:

Corollary 4.1. *(i) When $\theta \le \theta_0$, the threshold of fund opacity ω_0 is monotonically increasing in θ with $\lim_{\theta\to 0}\omega_0 = 0$.*

(ii) When $\theta > \theta_0$, both ω_L and ω_H are increasing in θ. Moreover, the region for multiple equilibria, (ω_L, ω_H), expands as θ increases.

For a given value of fund opacity ω_e, a weak flow-performance sensitivity supports only the low-β equilibrium, as the hiding motive strictly dominates the weak showing-off motive. However, as θ increases, a high flow-performance sensitivity encourages the manager to show off her skill by trading intensively. Hence, it gives birth to the high-β equilibrium, potentially leading to multiple equilibria.

Overall, the type of equilibrium depends on the combination of θ and ω_e. The tradeoff between the hiding and the showing-off motive tends to balance and generates multiple equilibria when the fund is moderately opaque and fund flows are sensitive to the expected investment performance. Otherwise, one of the conflicting motives dominates the other, making either the Kyle-like equilibrium or the flow-driven equilibrium the unique equilibrium.

Equilibrium comparison. Comparing the two stable equilibria, does the manager perform better in one of them? What are the implications for market liquidity, asset prices, and learning about asset-selection skills?

Corollary 4.2 (High-β vs. low-β equilibrium)**.** *In the high-β equilibrium, relative to the low-β equilibrium, the following results hold.*

(i) The market is more liquid, that is, λ is lower;

(ii) The manager's expected investment performance given the realized skill, $\mathrm{E}[(\delta - p)x|s] = \frac{\beta\omega_u s^2}{\beta^2\omega_s+\omega_u}$, is lower;

(iii) The price informativeness, defined by $\tau \equiv \frac{1}{\mathrm{Var}[\delta|p]} = \frac{\beta^2\omega_s+\omega_u}{\omega_\delta(\beta^2\omega_s+\omega_u)+\omega_u\omega_s}$*, is higher;*

(iv) The price volatility, $\mathrm{Var}[p] = \frac{\beta^2\omega_s^2}{\beta^2\omega_s+\omega_u}$*, is higher; and*

(v) Investors' estimate of the manager's skill is more precise, that is, $\hat{\omega}_s = \mathrm{Var}[s|f] = \frac{\omega_s\omega_u\omega_\delta}{\omega_s(\beta^2\omega_\delta+\omega_u)+\omega_u\omega_\delta}$ *is lower.*

Statement (i) of Corollary 4.2 holds because price impact λ is decreasing in β in the equilibrium (see footnote 16). Since the high-β equilibrium emerges only if the fund relatively transparent, the statement also implies that a higher fund transparency leads to higher market liquidity. The intuition is as follows. Relative to the low-β equilibrium, in the high-β equilibrium the manager trades more aggressively, and thus the order flow reflects significant information about the manager's skill s. To incorporate the appropriate amount of information about s into the price, the market maker seeks to offset the exaggerated information in order flow by lowering λ.

The intuition for statement (ii) is closely related to that for statement (i). In the high-β equilibrium, the manager earns a lower trading profit due to a smaller informational advantage over the market maker. This result may appear inconsistent with statement (i) because, ceteris paribus, a lower price impact is typically associated with a higher trading profit. However, the manager's order size $|x|$ in the high-β equilibrium is so large that the *overall* price reaction (i.e., λ multiplied by $|x|$) is larger than that in the low-β equilibrium. Consequently, the manager moves the price excessively and earns a lower trading profit in this equilibrium. Due to her aggressive trading, more information about s, which is informative about δ, is incorporated into the price in the high-β equilibrium, supporting statement (iii).

Statement (iv) follows, again, from the manager's aggressive trading in the high-β equilibrium. Specifically, she places a large buy (sell) order when $s > 0$ ($s < 0$), even if $|s|$ is small. This causes a large upward (downward) pressure on p, leading to greater price variability. This result aligns with the insight of Guerrieri and Kondor (2012) that reputation concerns amplify price volatility, although their mechanism differs from ours.[19]

In the high-β equilibrium, the manager reveals more information about s, thereby improving the precision of investors' estimation, as statement (v) demonstrates. This makes fund flows more responsive to skill, as investors can assess it more accurately.

4.4 Implications

This section discusses economic implications drawn from the equilibrium analyses under opacity.

4.4.1 Financial Fragility

The self-fulfilling nature of multiple equilibria suggests that financial markets may become vulnerable to non-fundamental factors. A shift in investor beliefs alone, without any structural changes, could cause the market to transition from the standard Kyle-like equilibrium to the flow-driven equilibrium, characterized by high price volatility and poor trading performance. We interpret this phenomenon as a form of *fragility* in financial markets.

Our model suggests that fragility may occur when the flow-performance sensitivity (θ) and opacity (ω_e) are moderately high. A high flow-performance sensitivity tends to arise when the fund management fee (ϕ) is relatively low. This surprising result goes counter to the recent debate on performance-based compesations for fund managers and advisory contracts, where payment structures that strongly reward performance are often criticized for encouraging excessive risk-taking. In our model, such performance-based compensation can actually stabilize the market by eliminating the flow-driven equilibrium and guiding the economy toward a unique low-β equilibrium. This is consistent with the empirical finding by Dass, Massa, and Patgiri (2008) that high-powered advisory contracts do not induce investment in bubbly stocks

[19] Guerrieri and Kondor (2012) argue that bond price volatility increases because fund managers demand a premium to offset the risk of reputation damage in the event of default.

but instead reduce such investments, highlighting the stabilizing effect of incentive compensations on financial markets.

4.4.2 Opacity and Trading Style

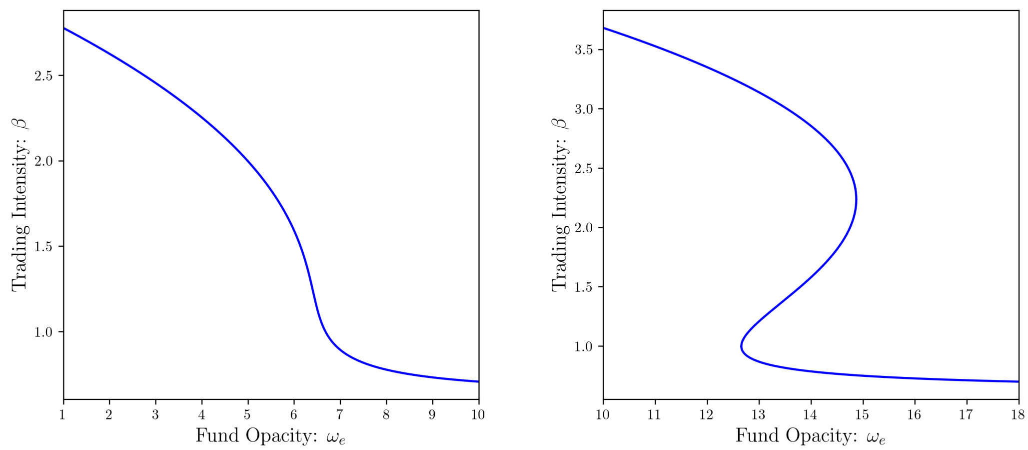
Furthermore, Proposition 4.1 yields novel implications connecting the fund opacity and its trading styles.

Corollary 4.3. *Both in the high- and low-β equilibria, the trading intensity β is monotonically decreasing in the fund opacity ω_e.*

Figure 4 illustrates the result in Corollary 4.3. First, as both panels illustrate, opaque funds tend to adopt low-intensity strategies that minimize market impact, prioritizing profits over fund flows. In contrast, transparent funds engage in high-intensity strategies with flashy risk taking, seeking reputation among investors and large fund flows. This relationship arises because the manager's reputation becomes insensitive to her trading behavior when the fund information is opaque and difficult to interpret for fund investors. The model predicts that these tendencies are more pronounced when the flow-performance sensitivity is weak, as the equilibrium is unique at such levels of θ (panel [a] of Figure 4).

These results are consistent with common observations in the real market. For example, quantitative hedge funds are often extremely opaque and highly profit-driven, focusing on consistent performance, with benefits from attracting fund flows being regarded as secondary to profitability (Schwager, 2012). On the other hand, some transparent funds, such as certain thematic ETFs or crypto-focused funds, engage in bold, attention-grabbing strategies to attract investor flows. These funds often emphasize visibility and narrative-driven positioning, which can lead to volatile returns and greater flow sensitivity.

The second implication is drawn from the model's multiple equilibria, as panel (b) in Figure 4 illustrates. Namely, the relation between funds' activities and their transparency may become blurry when the opacity is at an intermediate level. Notably, this indeterminacy may lead to financial fragility.



(A) Low Flow-performance Sensitivity

(B) High Flow-performance Sensitivity

Figure 4: Fund Opacity and Trading Intensity

Note: These figures are illustrated by seeting $\omega_s = 1.0, \omega_u = 2.0, \theta = 10(\phi = 0.08)$ for the left panel and $\theta = 27(\phi = 0.02)$ for the right panel.

Depending on market beliefs, relatively opaque funds may shift between profit-seeking and flow-driven strategies, creating non-fundamental fluctuations. Our result suggests that even relatively opaque funds may follow the crowd due to market beliefs, hopping between safe and risky strategies.

A notable observation in line with this result is momentum crashes (Daniel and Moskowitz, 2016), where funds engaging in risky and visible momentum trades often switch to safer strategies, generating sharp reversals and amplifying price declines. More generally, the rapid shifts between risk-taking and solid performance-based behavior have contributed to market fragility and volatility, as documented in various studies on economic crises, such as Brunnermeier (2009) on the credit crunch during the Global Financial Crisis in 2007. Importantly, large fundamental shifts that support such transitions are rarely identified (Gennotte and Leland, 1999). Our model does not require such fundamental shocks, as a change in beliefs alone can trigger a shift.

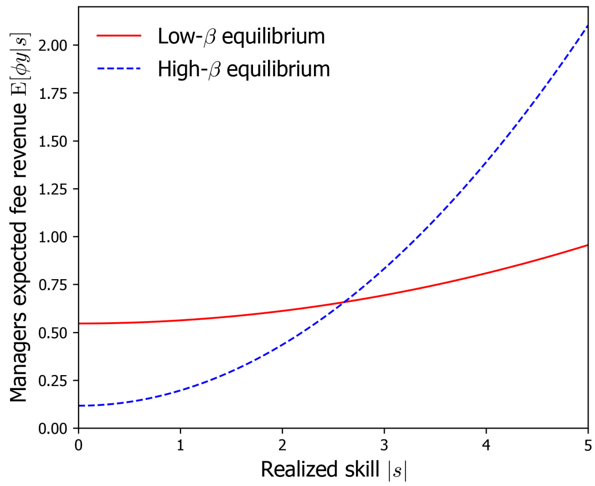


Figure 5: Skill versus Expected Wage

Note: The figure plots the manager's expected fee income at the low-β equilibrium (solid line) and high-β equilibrium (dashed line) for different levels of skill, $|s|$. The parameter values used in the figures are $\omega_s = 1$, $\omega_u = 2$, $\omega_e = 38$, and $\theta = 27$.

4.4.3 Compensation in Fund Management Industry

Our model provides insights into how skills are reflected in compensation in the fund management industry. As shown in statement (v) of Corollary 4.2, fund investors learn s more precisely in the high-β equilibrium than in the low-β equilibrium, as the manager demonstrates her skill by trading more intensively. Consequently, in the high-β equilibrium, the manager's expected compensation, $\mathrm{E}[\phi y|s]$, become more sensitive to her realized skill. As an illustration, Figure 5 plots the manager's expected compensation in relation to her skill in the low- and high-β equilibria. In the high-β equilibrium, investors' accurate evaluations result in greater pay inequality: low-skilled managers would receive very low compensation, while high-skilled ones are very highly paid. This result corroborates empirical observations, such as those provided by Philippon and Reshef (2012) and Böhm, Metzger, and Strömberg (2018), highlighting the comparatively large wage inequality in the fund management industry. Célérier and Vallée (2019) attribute its source to the scalability of skill, while inequality in our model arises from managers' incentives to attract fund flows through establishing high reputation.

4.4.4 Skill and Performance Persistence

Do high-skilled managers consistently perform well? Not always: skilled managers may not necessarily outperform less skilled ones. Consider a highly skilled manager and a relatively less skilled one. Since trading profits are increasing in managers' skill $|s|$, the former is expected to achieve higher trading profits than the latter *conditional on* both managers trading at the same intensity β. However, if their investors believe, for whatever reason, that the skilled manager invest more aggressively than the less skilled one, and managers conform to such beliefs, a high skill level does not translate into performance. This is because the skilled manager, trading at the flow-driven equilibrium, reveals too much private information to the market maker. Hence, compared to the less skilled manager, who plays the Kyle-like equilibrium, the skilled manager may earn lower expected profits. Figure 6 illustrates such a situation.[20] It plots the expected trading profits (performance) at high- and low-β equilibria with different levels of skill, suggesting that a low-skilled manager in the low-β equilibrium (point B) outperforms a high-skill manager in the high-β equilibrium (point A).

The empirical literature (e.g., Jensen, 1967; Busse, Goyal, and Wahal, 2010) finds that asset managers do not outperform passive benchmarks on average and, in cross-section, those who outperform cannot do so consistently. While the literature attributes the lack of persistent performance to the lack of skill, our result proposes an alternative explanation. Several studies reconcile the existence of skill with the lack of persistent performance. Berk and Green (2004), for example, argue that skill-based alphas eventually disappear due to competition between investors and decreasing returns to scale at the fund level.[21] Our model contributes to the understanding of this matter by showing

[20]The same argument holds for the comparison of the unconditional expected profit, $\mathrm{E}[\pi]$, using skill variance ω_s as an *ex-ante* measure of the skill level.

[21]Pastor, Stambaugh, and Taylor (2015) document increases in skill in finance and attribute the lack of corresponding increases in performance to decreasing returns to scale at the industry level.

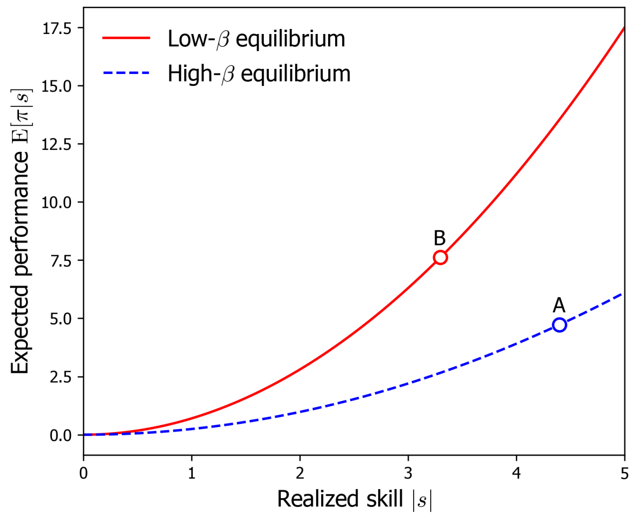


Figure 6: Skill versus Expected Trading Profit

Note: The figure plots the expected trading profit $\mathrm{E}[\pi]$ at the low-β equilibrium (solid line) and the high-β equilibrium (dashed line) for different levels of skill $|s|$. The parameter values used in the figure are $\omega_s = 1$, $\omega_u = 2$, $\omega_e = 38$, and $\phi = 0.02$.

that the lack of interplay between manager performance and skill could be the result of self-fulfilling multiple equilibria.

5 Strategic Opacity

This section analyzes endogenous opacity by allowing the manager to control ω_e. It uncovers the manager's incentive to intentionally make the fund opaque and how it varies depending on her expected skill.

5.1 Setup

Prior to investors' capital allocation and the manager's investment decision, the fund manager chooses the fund opacity ω_e, anticipating the equilibrium reactions of investors and the market maker specified in Sections 3 and 4. The manager seeks to maximize her *ex-ante* expected utility derived from the fee income, that is, $U \equiv \mathrm{E}[\phi y]$. By incorporating the fund flow in equation (3.6), her objective function is specified as follows.[22]

$$U = \phi \mathrm{E}[\pi + M] = \phi(1+\theta)\mathrm{E}[\pi] - c(\omega_e). \tag{5.1}$$

To formalize the manager's optimization problem in the rational expectations framework, we assume that, when parameter values support multiple equilibria, all agents in the economy coordinate their beliefs toward the low-β equilibrium.[23] Also, for the sake of tractability, we assume that $c'(\omega_e)$ is sufficiently large, so that the manager's maximization problem satisfies the second-order condition.

In this extended model, the definition of equilibrium augments Definition 1 of the baseline model by additionally requiring that ω_e maximizes the manager's objective function U in equation (5.1).

5.2 Opacity as Commitment Device

Given the trading intensity β, the unconditional expected trading profit in (5.1) is computed as

$$\mathrm{E}[\pi] = \frac{\beta\omega_u}{\beta^2\omega_u + \omega_s}\omega_s. \tag{5.2}$$

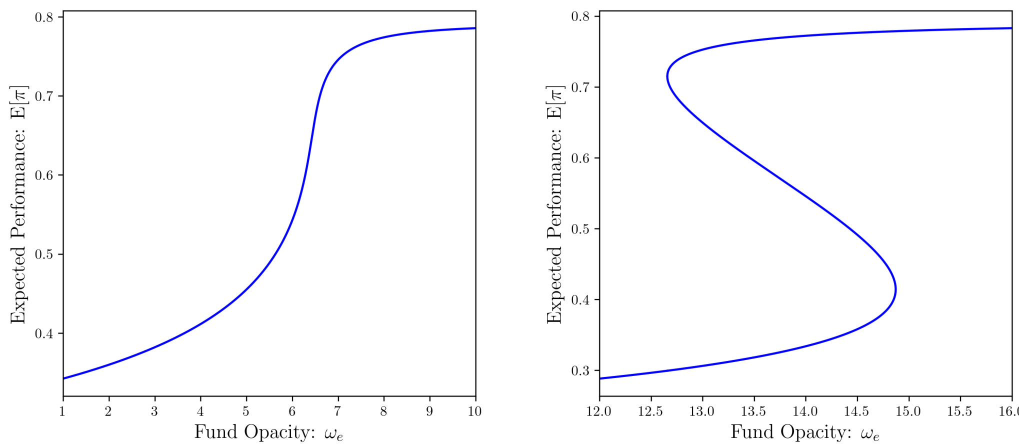
As in the standard Kyle (1985) framework, the first fraction represents the profit margin for each unit of the informational advantage of the manager, and the second term ω_s represents the expected size of this advantage.

The fund opacity does not directly show up in equation (5.2), but it influences the expected performance through the equilibrium trading intensity β.

Corollary 5.1. *The expected fund performance, defined by* $\mathrm{E}[\pi]$*, is monotonically increasing in the fund opacity.*

[22] The law of iterated expectation implies $\mathrm{E}\left[\mathrm{E}[\pi|f]\right] = \mathrm{E}[\pi]$.

[23] Alternatively, we may assume that the agents rely on the realization of a sunspot shock to coordinate their beliefs, where they anticipate the low- and the high-β equilibria with probability $\rho \in [0,1]$ and $1-\rho$, respectively. Our setting corresponds to $\rho = 1$, and this assumption does not affect our qualitative results.



(A) Low Flow-performance Sensitivity (B) High Flow-performance Sensitivity

Figure 7: Fund Opacity and Expected Performance

Note: The figure plots the unconditional expected performance of the fund $\mathrm{E}[\pi]$ against fund opacity ω_e with different values of the flow-performance sensitivity: $\theta = 10$ (the left panel) and $\theta = 27$ (the right panel). The value of other parameters are $\omega_s = 1.5$ and $\omega_u = 3$.

Figure 7 plots $\mathrm{E}[\pi]$ in relation to the fund opacity ω_e, which traces the reactions of β in Figure 4. In our model, $\mathrm{E}[\pi]$ declines with the trading intensity β. This is due to the presence of the showing-off motive: althrough the unconditional expected performance is maximized at the Kyle (1985)'s benchmark with $\theta = 0$, the manager seeks to trade more aggressively to attract fund flows. It revelas too much private information to the market maker and lowers the fund performance, while the manager is compensated for this loss by attracting more fund flows. The opacity, conversely, diminishes the showing-off motive and induces the manager to take more profit-driven strategies, bringing β closer to the profit-maximizing level. As a consequence, opacity helps improve the expected fund performance and, according to the manager's utility (5.1), this effect encourages the manager to make her fund opaque.

However, given that fund flow M increases with the expected performance (see equation [3.6]), the showing-off motive may seem paradoxical: why does the manager trade aggressively to begin with, knowing it harms performance and reduces M? This tension arises from a crucial information timing gap. When she decides on the trading strategy x, the manager cannot alter the market's beliefs about her strategy (β and λ in equations [4.1]–[4.4]), but she can influence investor perceptions via the signal f by inflating her trading volume $|x|$ to appear more skilled, creating the showing-off motive. Conversely, at the *ex-ante* stage (i.e., before she sends the signal f to investors), the manager recognizes that this future showing-off motive will lead to flashy but inefficient risk-taking that erodes unconditional performance and the expected fund flow $\mathrm{E}[M]$. Consequently, she strategically chooses fund opacity as a commitment device to "tie her own hands," thereby dampening the investors' future responsiveness to f and forcing her future self to prioritize profit-driven trading over reputation-driven trading.

5.3 Implications

5.3.1 Skill, Opacity, and Fragility

We first propose "middling" skill acquisition due to the multiple trading-stage equilibria by focusing on $\omega_s \in (\underline{\omega}_s, \bar{\omega}_s)$. We refer to it as middling because the acquired skill level is either higher or lower than what she would acquire if she knew which type of equilibrium would be realized. On the equilibrium path, both the selected skill potential (ω_s^*) and the market's belief about it (ω_s) fall in this region if the marginal skill-acquisition cost is intermediate. In this region, the trading stage involves multiple equilibria (i.e., $\beta_0 = \beta_H$ or β_L), whereas the trader at the skill-acquisition stage cannot influence the equilibrium selection. Consequently, she determines her skill potential by anticipating the sunspot shock, which leads to the skill acquisition as follows.

Proposition 5.1 (Middling skill acquisition)**.** *The equilibrium skill potential in the presence of multiple trading-stage equilibria is*

(i) lower than the skill potential that she would acquire in the low-β equilibrium; and

(ii) higher than the skill potential that she would acquire in the high-β equilibrium.

Figure **??** illustrates a numerical example; $\omega_{s,m}$ is the ex-ante optimal skill potential, while the ex-post equilibrium is either $\omega_{s,\ell}$ or $\omega_{s,h}$, depending on the equilibrium realization in the trading stage.

The trader invests in skill potential incorporating the possibility of both high- and low-β equilibria. At the trading stage, however, the acquired skill potential turns out to be insufficient if the low-β equilibrium is realized, as its optimal play involves the profit-driven strategy and requires a relatively high asset-selection skill (i.e., $\omega_{s,h}$ is the *ex-post* equilibrium). Conversely, the acquired skill potential is too high if the high-β equilibrium is realized, as the trader takes the reputation-driven strategy that does not require a high skill potential (i.e., $\omega_{s,\ell}$ is the *ex-post* equilibrium).

This result indicates that individuals or firms may invest in skills that seem oddly intermediate or cautious—neither fully specialized nor overly basic. For example, it explains why some traders and finance professionals choose to pursue broader business qualifications instead of more specialized academic paths, allowing them to adapt to a wide range of market situations. Traders also learn intermediate-level coding (e.g., Python) for algorithmic trading or data analysis but stop short of deep computer science expertise. A similar trend can be observed with respect to CFA Levels and quantitative skills among hedge fund managers. These skill acquisitions would be difficult to rationalize in an economy with a single equilibrium but become logical under the equilibrium multiplicity and uncertainty regarding equilibrium realization.

5.3.2 Fragile Skill Acquisition and Skill Diversity

By expanding our scope to the entire range of skill potential (ω_s), the discontinuities in Lemma **??** imply that the skill-acquisition problem in (**??**) may have multiple solutions.[24]

For example, when the marginal cost of skill acquisition is moderately small, as Figure **??** illustrates, both the high skill potential ($\omega_{s,h}$) and the in-

[24] Although the equilibrium multiplicity in the trading stage (i.e., β_0) is determined by the level of ω_s, as Proposition **??** stipulates, it is orthogonal to the existence of multiple solutions to the optimal skill acquisition in equation (**??**).

termediate skill potential ($\omega_{s,m}$) become optimal for the trader. Corresponding to $\omega_{s,h}$, the equilibrium in the trading stage is unique with $\beta_0 = \beta_L$, where the trader engages in solid profit-driven trading. Conversely, the trading stage with skill potential $\omega_{s,m}$ involves multiple equilibria, and the selected skill potential is middling, as Proposition 5.1 demonstrates. The symmetric argument holds when the marginal cost is relatively high: the intermediate skill potential and the low skill potential show up as multiple optima. This result can explain at least three realistic phenomena without relying on ad-hoc assumptions on the skill-acquisition environment.

Firstly, the coexistence of traders with different skill levels arises not from heterogeneity in skill-acquisition problems (e.g., different marginal costs), but from the multiple trading-stage equilibria and the possibility of mixed strategies in the skill acquisition stage. This reflects a realistic scenario where, due to advancements in the internet and social media (e.g., online webinars via Zoom and academic courses uploaded on YouTube), opportunities for skill acquisition have become widely and uniformly accessible to all traders, reducing barriers to learning. Despite the increased homogeneity in skill-acquisition costs, we continue to observe heterogeneity in skill levels in the finance industry. Our model offers an explanation for this observation, showing how diverse skill levels can emerge among market participants, without the need to assume inherent differences in their skill acquisition costs.

Secondly, it also captures the possibility of widespread surges in skill acquisition without assuming special conditions for the skill-acquisition environment. For example, we can explain traders and fund managers in competitive environments (e.g., Wall Street) may collectively strive to acquire high-level skills, such as pursuing advanced degrees, driven by the mixed-strategy equilibrium. This surge in skill acquisition could occur even in the absence of significant changes in external factors, and without the need for heterogeneity in skill-acquisition costs, reflecting industry-wide trends in upskilling and degree inflation, as documented by Philippon and Reshef (2012) and Fuller and Raman (2017).

Finally, discontinuities in the marginal benefit curve add another layer of insight into the effect of the skill-acquisition cost on the equilibrium skill potential. Namely, discontinuities introduce the possibility of significant jumps in the optimal skill level triggered by a small change in the marginal skill-acquisition cost. For instance, when marginal technological advancements slightly reduce the marginal cost of learning, the result could be a sizable upward shift in skill acquisition. Conversely, even a small increase in the cost of learning might cause a large decline in the equilibrium skill level. This discontinuity-driven sensitivity can help explain the historical observations stylized by Philippon and Reshef (2012) that certain external changes lead to abrupt shifts in the distribution of skill levels in the finance industry.

6 Conclusion

This paper studies a model à la Kyle (1985) where traders are privately informed about their skill, i.e., the trader-specific ability to identify potentially mispriced assets. The traders care not only about trading profits but also the recruiter's perception of their skill. The model has two stable self-fulfilling equilibria, which we interpret as fragility in financial markets. One of the equilibria is similar to that of Kyle (1985) where reputation is irrelevant. In the other, "reputation-driven" equilibrium, the traders trade more aggressively for fear of being seen as unskilled by the recruiter, causing higher price volatility, higher market liquidity, and lower trading profits. Our model predicts that fragility may arise when poaching is (moderately) active in the finance industry and when there are high demands for financial skill. The model yields implications connecting traders' skills, their trading styles, and financial fragility. Moreover, consistent with empirical observations, our model predicts large pay inequality in the finance industry and reconciles skill in active management and the lack of performance persistence.

As a direction for future research, it would be interesting to analyze various types of wage structures in financial markets, ranging from back-end to front-end compensations. The implications of the forward-looking trading motives introduced in our model can be applied to investigate which types of wage structures and employment systems can mitigate financial market fragility while also improving the welfare of market participants.

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Online Appendix

A Proof of Lemma ??

From (3.1) and (3.2), p is observationally equivalent to $\frac{p-\hat{\delta}}{\lambda} = \beta s + u \equiv z_p$, which is the sum of two normally distributed variables from the market maker's perspective. Since $\delta \sim N(\hat{\delta} + s, \omega_\delta)$ is equivalently presented as $\delta = \hat{\delta} + s + \epsilon$ with $\epsilon \sim N(0, \omega_\delta)$, δ is observationally equivalent to $\delta - \hat{\delta} = s + \epsilon \equiv z_\delta$, which is also the sum of normal variables. Thus, given $Z = [z_p,\ z_\delta]^{\mathrm{T}}$, the market maker's posterior mean of s is

$$\mathrm{E}\left[s|Z\right] = \underbrace{\mathrm{E}[s]}_{=0} + \Sigma_{sZ}\Sigma_Z^{-1}(Z - \underbrace{E[Z]}_{=[0\ \ 0]^{\mathrm{T}}}), \tag{A.1}$$

where

$$\Sigma_{sZ} = \begin{bmatrix} \beta\omega_s, & \omega_s \end{bmatrix} \text{ and } \Sigma_Z = \begin{bmatrix} \beta^2\omega_s + \omega_u, & \beta\omega_s \\ \beta\omega_s, & \omega_s + \omega_\delta \end{bmatrix}. \tag{A.2}$$

Since

$$\begin{aligned} \det[\Sigma_Z] &= (\beta^2\omega_s + \omega_u)(\omega_s + \omega_\delta) - \beta^2\omega_s^2 \\ &= \omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta, \end{aligned} \tag{A.3}$$

it holds that

$$\begin{aligned} \mathrm{E}\left[s|Z\right] &= \begin{bmatrix} \beta\omega_s, & \omega_s \end{bmatrix} \frac{1}{\det[\Sigma_Z]} \begin{bmatrix} \omega_s + \omega_\delta, & -\beta\omega_s \\ -\beta\omega_s, & \beta^2\omega_s + \omega_u \end{bmatrix} \begin{bmatrix} z_p \\ z_\delta \end{bmatrix} \\ &= \frac{\omega_s}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \left(\beta\omega_\delta z_p + \omega_u z_\delta\right) \\ &= \frac{\omega_s(\beta\omega_\delta + \omega_u)}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \left(\frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u} z_p + \left(1 - \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u}\right) z_\delta \right) \\ &= \xi\left(\gamma z_p + (1-\gamma) z_\delta\right), \end{aligned} \tag{A.4}$$

where

$$\xi \equiv \frac{\omega_s(\beta\omega_\delta + \omega_u)}{\omega_s(\beta^2\omega_\delta + \omega_u) + \omega_u\omega_\delta} \tag{A.5}$$

and

$$\gamma \equiv \frac{\beta\omega_\delta}{\beta\omega_\delta + \omega_u}. \tag{A.6}$$

Re-writing z_p and z_δ in (A.4) in terms of p and δ, we obtain the required result. $\square$

B Proof of Lemma ??

Let us present $\delta \sim N(\hat{\delta} + s, \omega_\delta)$ equivalently as $\delta = \hat{\delta} + s + \epsilon$ with $\epsilon \sim N(0, \omega_\delta)$. Then, squaring (**??**), we have

$$\begin{aligned} \hat{s}^2 &= \xi^2 \left(\gamma^2 x^2 + 2\gamma x \left(\gamma u + (1-\gamma)(\delta - \hat{\delta}) \right) + \left(\gamma u + (1-\gamma)(\delta - \hat{\delta}) \right)^2 \right) \\ &= \xi^2 \left(\begin{array}{c} \gamma^2 x^2 + 2\gamma^2 x u + 2\gamma(1-\gamma)(s+\epsilon)x \\ + \gamma^2 u^2 + 2\gamma(1-\gamma)(s+\epsilon)u + (1-\gamma)^2(s+\epsilon)^2 \end{array} \right). \end{aligned} \tag{B.1}$$

Taking the expectation of (B.1) from the trader's perspective,

$$\mathrm{E}[\hat{s}^2|s] = \xi^2 \left(\gamma^2 x^2 + 2\gamma(1-\gamma)sx + A \right), \tag{B.2}$$

where

$$A \equiv \gamma^2 \omega_u + (1-\gamma)^2 (s^2 + \omega_\delta) \tag{B.3}$$

is the term that the trader is unable to influence by her choice of x. We can omit A in the maximization problem below. Using (B.2), the trader's problem (**??**) is rewritten as

$$\begin{aligned} &\max_x \ \mathrm{E}\left[(\delta - p)x|s\right] + \theta \mathrm{E}[\hat{s}^2|s] \\ \Rightarrow \ &\max_x \ \mathrm{E}\left[\left(\hat{\delta} + s + \epsilon - \left(\hat{\delta} + \lambda(x+u) \right) \right) x \middle| s \right] + \theta \mathrm{E}[\hat{s}^2|s] \\ \Rightarrow &\max_x \ sx - \lambda x^2 + \theta \xi^2 \left(\gamma^2 x^2 + 2\gamma(1-\gamma)sx \right). \quad \square \end{aligned} \tag{B.4}$$

C Proof of Lemma ??

At the beginning of $t = 1$, the trader does not have reputation concerns anymore. Her problem is just to choose x_1 to maximize the expected trading profit, that is,

$$\max_{x_1 \in (-\infty, \infty)} \mathrm{E}[(\delta_1 - p_1)x_1 | s]. \tag{C.1}$$

We conjecture and later verify that the trader's equilibrium order in $t = 1$ is

$$x_1 = \beta_1 s, \tag{C.2}$$

where $\beta_1 > 0$ is a constant to be determined. Given (C.2), the market maker sets the price

$$\begin{aligned} p_1 =& \mathrm{E}\left[\delta_1 | x_1 + u_1\right] \\ =& \mathrm{E}[\mathrm{E}[\delta_1 | x_1 + u_1, s] | x_1 + u_1] \\ =& \mathrm{E}[\hat{\delta}_1 + s | x_1 + u_1] \\ =& \hat{\delta}_1 + \frac{\mathrm{Cov}[s, x_1 + u_1]}{\mathrm{Var}[x_1 + u_1]} \left(x_1 + u_1 - \mathrm{E}[x_1 + u_1]\right) \\ =& \hat{\delta}_1 + \frac{\mathrm{Cov}[s, \beta_1 s + u_1]}{\mathrm{Var}[\beta_1 s + u_1]} \left(x_1 + u_1 - \mathrm{E}[\beta_1 s + u_1]\right) \\ =& \hat{\delta}_1 + \lambda_1 \left(x_1 + u_1\right), \end{aligned} \tag{C.3}$$

where

$$\lambda_1 \equiv \frac{\beta_1 \omega_s}{\beta_1^2 \omega_s + \omega_{u,1}}. \tag{C.4}$$

Given (C.3), the trader's problem (C.1) is rewritten as

$$\max_{x_1} \; \mathrm{E}\left[\left(\delta_1 - \left(\hat{\delta}_1 + \lambda_1(x_1 + u_1)\right)\right) x_1 \middle| s\right] \; \Rightarrow \; \max_{x_1} \; s x_1 - \lambda_1 x_1^2. \tag{C.5}$$

The FOC for x_1 is $s - 2\lambda_1 x_1 = 0$, which determines the trader's optimal action:

$$x_1 = \frac{1}{2\lambda_1} s. \tag{C.6}$$

For the other agents' belief (C.2) to be consistent with (C.6), we need

$$\beta_1 = \frac{1}{2\lambda_1} \iff \beta_1 = \sqrt{\frac{\omega_{u,1}}{\omega_s}}, \tag{C.7}$$

which implies

$$\lambda_1 = \frac{1}{2}\sqrt{\frac{\omega_s}{\omega_{u,1}}}. \quad \square \tag{C.8}$$

D Proof of Lemma ??

Let us present $\delta_1 \sim N(\hat{\delta}_1 + s, \omega_\delta)$ equivalently as $\delta_1 = \hat{\delta}_1 + s + \epsilon_1$ with $\epsilon_1 \sim N(0, \omega_\delta)$. Then the trader's trading profit in $t = 1$ is

$$\begin{aligned}
\pi_1 &= (\delta_1 - p_1)x_1 \\
&= \left(\hat{\delta}_1 + s + \epsilon_1 - \left(\hat{\delta}_1 + \lambda_1\left(\beta_1 s + u_1\right)\right)\right)\beta_1 s \\
&= \left(\hat{\delta}_1 + s + \epsilon_1 - \left(\hat{\delta}_1 + \frac{1}{2}\sqrt{\frac{\omega_s}{\omega_{u,1}}}\left(\sqrt{\frac{\omega_{u,1}}{\omega_s}}s + u_1\right)\right)\right)\sqrt{\frac{\omega_{u,1}}{\omega_s}}s \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}s^2 + \left(\sqrt{\frac{\omega_{u,1}}{\omega_s}}\epsilon_1 - \frac{u_1}{2}\right)s. \qquad \text{(D.1)}
\end{aligned}$$

Taking the expectation of (D.1) from the recruiter's perspective, we obtain the trader's salary:

$$\begin{aligned}
w &= \mathrm{E}[\pi_1 | p_0, \delta_0] \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\mathrm{E}[s^2 | p_0, \delta_0] \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\left(\mathrm{E}[s|p_0, \delta_0]^2 + \mathrm{Var}[s|p_0, \delta_0]\right) \\
&= \frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}\left(\hat{s}^2 + \hat{\omega}_s\right), \qquad \text{(D.2)}
\end{aligned}$$

where $\hat{\omega}_s$ is the posterior variance of s, which is a constant. $\square$

E Proof of Propositions 4.1 and ??

In the following analyses, denote

$$a = \frac{\omega_{u,0}}{\omega_s},\ b = \frac{\omega_{u,0}}{\omega_\delta}, \tag{E.1}$$

so that we rewrite equation (4.8) as

$$\beta = F(\beta) \equiv \frac{1 + 2\theta \left(\frac{1}{a+b+\beta^2}\right)^2 b\beta}{2\left(\frac{\beta}{\beta^2+a} - \theta\left(\frac{\beta}{a+b+\beta^2}\right)^2\right)}. \tag{E.2}$$

By rearranging (E.2), the equilibrium β solves the following condition.

$$2\theta\beta = J(\beta^2), \tag{E.3}$$

where, by denoting $\beta^2 = y$,

$$J(y) = \frac{(y-a)(y+b+a)^2}{(y+a)(y+b)}. \tag{E.4}$$

The first-order derivative of the RHS of (E.3) with respect to β is

$$K^{(1)}(\beta) \equiv 2\beta J^{(1)}(\beta^2), \tag{E.5}$$

with

$$J^{(1)}(y) \equiv \frac{N(y)}{(y+a)^2(y+b)^2}, \tag{E.6}$$

where the numerator is

$$N(y) = y^4 + 2(a+b)y^3 + n_2 y^2 + n_1 y + n_0$$

with coefficients being

$$n_2 = 2a^2 + 6ab + b^2,$$

$$n_1 = 2a\left[3b^2 + 3ab + a^2\right],$$

$$n_0 = a\left(a^2+ab+2b^2\right)(a+b).$$

From (E.4) and (E.6), $y=a$ is the unique solution to $J(y)=0$ in $y>0$, and $J(y)$ is increasing in y for $y \geq a$. Hence, there is no equilibrium in $y<a$, and we focus on $y \geq a$.

The second-order derivative of the RHS of (E.3) with respect to β is expressed as

$$K^{(2)}(y) \equiv 2\left(J^{(1)}(y)+2yJ^{(2)}(y)\right), \tag{E.7}$$

where $J^{(2)}$ denotes the second-order derivative of J with respect to y. $K^{(1)}$ and $K^{(2)}$ have the following properties:

Lemma E.1. *(i) $K^{(1)}(\beta)$ takes a U-shaped curve in relation to β.*
(ii) $K^{(2)}(a)<0$ and $K^{(2)}(\infty)>0$. There exists a unique solution to $K^{(2)}(y)=0$.

Proof. Note that point (i) directly follows from point (ii). By denoting that $N^{(1)}(y)=\frac{dN(y)}{dy}$, it holds that

$$J^{(1)}(y)+2yJ^{(2)}(y)=J^{(1)}(y)\frac{-7y^2-3(a+b)y+ab}{(y+a)(y+b)}+2y\frac{\left(y^2+(a+b)y+ab\right)N^{(1)}(y)}{(y+a)^3(y+b)^3}. \tag{E.8}$$

Evaluating at $y=a$,

$$\begin{aligned}
J^{(1)}(a)+2aJ^{(2)}(a) &= -J^{(1)}(a)\frac{5a+b}{a+b}+\frac{N^{(1)}(y)}{2a(a+b)^2} \\
&\propto -2a(a+b)(5a+b)J^{(1)}(a)+N^{(1)}(a) \\
&= -4a^3-b(4+b)a-b^3 \\
&< 0.
\end{aligned}$$

Therefore, $K^{(2)}(a)<0$ holds.

The numerator of (E.8) is summarized as

$$
\begin{aligned}
T(y) = &- \left[y^4 + 2(a+b)y^3 + n_2 y^2 + n_1 y + n_0\right] \left[7y^2 + 3(a+b)y - ab\right] \\
&+ 2y \left(y^2 + (a+b)y + ab\right) \left[4y^3 + 6(a+b)y^2 + 2n_2 y + n_1\right].
\end{aligned}
$$

The above argument suggests that $T(a) < 0$.

Using Mathematica, we summarize T into the following polynomial:

$$T(y) = \sum_{k=0}^{6} t_k y^k, \tag{E.9}$$

where the coefficients are

$$t_0 = a^2 b(a+b)\left(a^2 + ab + 2b^2\right) > 0,$$

$$t_1 = -3a\left(a^4 + (a+b)(a-b)^2 b + b^4\right) < 0,$$

$$t_2 = -3a\left(3a^3 + 4a^2 b + ab^2 + 5b^3\right) < 0,$$

$$t_3 = -8a^3 - 8a^2 b - 9ab^2 + b^3,$$

$$t_4 = 3b\,(a+b) > 0,$$

$$t_5 = 3(a+b) > 0,$$

and $t_6 = 1$. As $t_6 > 0$, $T(\infty) = \infty > 0$. Together with $T(a) < 0$, the intermediate value theorem implies that $T(y) = 0$ has at least one solution in $y > a$.

Suppose that y_{min} is one of the solutions to $T(y) = 0$. Consider the first-order derivative of T evaluated at $y = y_{min}$ and multiplied y_{min}:

$$\mathcal{T}(y_{min}) = y_{min} \frac{dT(y_{min})}{dy} = \sum_{k=1}^{6} k t_k y_{min}^k.$$

Applying $T(y_{min}) = 0$ and eliminating the terms related to t_3,

$$\mathcal{T}(y_{min}) = -3t_0 - 2t_1 y_{min} - t_2 y_{min}^2 + t_4 y_{min}^4 + 2t_5 y_{min}^5 + 3t_6 y_{min}^6,$$

According to the signs of t_k and $y_{min} > a$, it holds that

$$\begin{aligned}\mathcal{T}(y_{min}) &> -3t_0 - 2t_1 a - t_2 a^2 + t_4 a^4 + 2t_5 a^5 + 3t_6 a^6 \\ &= 3a^2\left(6a^4 + 8a^3 b + b^4 + (a^2 - b^2)^2\right) \\ &> 0.\end{aligned}$$

Therefore, if y_{min} is a solution to $T(y) = 0$, then the slope of $T(y)$ at this point must be positive. Together with $T(a) < 0 < T(\infty)$, y_{min} is the unique solution to $T(y) = 0$, and $T(y) > 0 \Leftrightarrow y > y_{min}$.[25] The argument above implies that $K^{(1)}(\beta)$ takes a U-shaped curve with respect to β, and the minimum point is given by $\beta_{min} \equiv \sqrt{y_{min}}$. Note that $\tilde{\beta}$ in footnote **??** is given by $\tilde{\beta} = \beta_{min}$. $\square$

The RHS of (E.3) is monotonically increasing in β, and Lemma E.1 attests that it is concave in $y \leq y_{min}$, while it becomes convex in $y > y_{min}$. In other words, the slope of the curve is the least steep at $y = y_{min}$, and it is given by

$$K_{min} = 2\beta_{min} J^{(1)}(y_{min}). \tag{E.10}$$

Note that K_{min} is represented only by parameters, a, b, and does not involve θ.

Consider the following condition to characterize equilibrium:

$$J(y_{min}) > K_{min}\beta_{min}. \tag{E.11}$$

Then the following result holds:

Lemma E.2. *(i) If inequality (E.11) holds, then there exist θ_H and θ_L with $\theta_H > \theta_L$. If, and only if, $\theta \in \Theta \equiv (\theta_L, \theta_H)$, then the equilibrium condition (E.3) has three solutions.*
(ii) If inequality (E.11) is violated, then $\theta_H = \theta_L$, and the equilibrium condition (E.3) has a unique solution. The solution is $\beta > \beta_{min}$ ($\beta < \beta_{min}$) if $\theta > \theta_H$

[25]Suppose that $T(y) = 0$ has more than two solutions. For any solutions, denoted as y_0 and y_1, the mean-value theorem implies that there exists at least one $y_2 \in [y_0, y_1]$ such that $\frac{dT(y_2)}{dy} = 0$. This contradicts to the strict positivity or negativity of $\mathcal{T}$.

($\theta < \theta_H$).

Proof. Consider the plot of the curve, $J(\beta^2)$, and the line, $2\theta\beta$, against β. With the curve being fixed, consider a line that goes through the origin and is tangent to the curve. The tangent point is characterized by β that satisfies

$$J(\beta^2) = K^{(1)}(\beta^2)\beta,$$

where the LHS represents the y-coordinate of the curve at β, and the RHS represents that of the line that goes through the origin. Note that their slopes must be identical and $K^{(1)}(\beta^2)$. Applying (E.5), the above condition is rewritten as

$$J(\beta^2) = 2\beta^2 J^{(1)}(\beta^2).$$

Hence, denoting $y = \beta^2$, the tangent point is characterized by the solution to

$$0 = G(y) = 2yJ^{(1)}(y) - J(y).$$

Note that we focus on $y > \sqrt{a}$, and thus y and β have one-to-one mapping.

Function $G(y)$ has the following first-order derivative:

$$\begin{aligned} G^{(1)}(y) &= 2\left(yJ^{(2)}(y) + J^{(1)}(y)\right) - J^{(1)}(y) \\ &= J^{(1)}(y) + 2yJ^{(2)}(y) \\ &= \frac{K^{(2)}(y)}{2}, \end{aligned}$$

where we apply the definition of $K^{(2)}$ in (E.7). Lemma E.1 has established that $K^{(2)}(a) < 0$ and $K^{(2)}(\infty) > 0$. Also, there is a unique $y = y_{min}$ such that $K^{(2)}(y_{min}) = 0$ and $K^{(2)}(y_{min}) > 0 \Leftrightarrow y > y_{min}$. Hence, $G(y)$ takes a U-shaped curve with respect to y and $y = y_{min}$ is the minimizer of the curve.

With $G(y)$ being U-shaped, $G(y) = 0$ has two solutions if and only if its minimum point dips below the x-axis, namely,

$$2y_{min}J^{(1)}(y_{min}) - J(y_{min}) < 0.$$

Applying the definition of K_{min} in (E.10), the above inequality is rewritten as condition (E.11).

When condition (E.11) holds and $G(y) = 0$ has two solutions, we denote them as y_0 and $y_1 (> y_0)$. The corresponding slopes of these tangent lines are $K^{(1)}(y_0)$ and $K^{(1)}(y_1)$, respectively. Then the original line, $2\theta\beta$, and the curve, $J(\beta^2)$, have three intersections if and only if

$$\frac{K^{(1)}(y_1)}{2} \equiv \theta_L < \theta < \theta_H \equiv \frac{K^{(1)}(y_0)}{2}. \tag{E.12}$$

This result concludes the proof of statement (i) of the lemma.

Conversely, if condition (E.11) does not hold, no tangent line that goes through the origin is obtained: function $J(\beta^2)$ becomes the least steep at $\beta_{min} = \sqrt{y_{min}}$, and the tangent line is specified by $K_{min}\beta - K_{min}\beta_{min} + J(\beta_{min}^2)$, while the intercept of this line is negative, as condition (E.11) is violated. Hence, the line, $2\theta\beta$, and the curve, $J(\beta^2)$ have a unique intersection. This intersection occurs in $\beta > \beta_{min}$ if, and only if, $\theta > \theta'_H \equiv \frac{J(\beta_{min}^2)}{2\beta_{min}}$.

To conclude statement (ii), we must show that θ_H and θ_L in statement (i) converges to θ'_H in statement (ii) at the threshold of these two cases so that we re-define it as $\theta_H = \theta_L = \theta'_H$. This result is shown in Lemma E.3 below. □

Finally, consider the comparative statics with respect to ω_s. The following result holds:

Lemma E.3. *(i) $J(y_{min}) - K_{min}\beta_{min}$ is increasing in ω_s. Hence, there exists a unique $\tilde{\omega}_s > 0$, such that, (E.11) holds with equality, and inequality (E.11) holds if and only if $\omega_s > \tilde{\omega}_s$.*
(ii) When condition (E.11) holds, both θ_H and θ_L increase as ω_s increases. Interval Θ expands, as the following inequalities hold:

$$0 < \frac{d\theta_L}{d\omega_s} < \frac{d\theta_H}{d\omega_s}.$$

(iii) At $\omega_s = \tilde{\omega}_s$, it holds that $\theta_H = \theta_L = \theta'_H$, while at the limit of $\omega_s \to 0$, $\theta'_H \to 0$.

(iv) There exists a unique threshold for φ^*, such that, $\varphi \geq \varphi^*(< \varphi^*)$ leads to case 1 (case 2) in Proposition **??**.

Proof of Statements (i) and (ii). Note that we are interested in the reaction of the following function when $a = \frac{\omega_{u,0}}{\omega_s}$ changes and, hereafter, we make the dependence of functions on a explicit:

$$2\theta_k = K^{(1)}(y_j, a) = 2\sqrt{y_j} J^{(1)}(y_j, a),$$

where $y_j = y_0, y_1$ denote the solutions to $G(y, a) = 0$, if any, and $k = L, H$ are appropriately determined following (E.12). According to the definition of G, the following partial derivative holds.

$$\frac{\partial G(y, a)}{\partial a} = 2y \frac{\partial J^{(1)}(y, a)}{\partial a} - \frac{\partial J(y, a)}{\partial a}.$$

The first term is summarized by

$$\begin{aligned}
\frac{\partial J^{(1)}(y, a)}{\partial a} &\propto (y + a) \frac{\partial N(y, a)}{\partial a} - 2N(y, a) \\
&= (y + a)\left[2y^3 + n_{2,a}y^2 + n_{1,a}y + n_{0,a}\right] - \left[y^4 + 2(a + b)y^3 + n_2 y^2 + n_1 y + n_0\right] \\
&= \sum_{k=0}^{4} A_k y^k
\end{aligned}$$

where $A_4 = 1$, $A_3 = 4(a + b)$,

$$A_2 = 2a^2 + 5b^2 + 2(3ab + a^2) + 2a\left[3b + 2a\right],$$

$$A_1 = 2a^2\left[3b + 2a\right]\left(a^2 + ab + 2b^2\right)(2a + b) + a\,(2a + b)\,(a + b),$$

$$\begin{aligned}
A_0 &= a\left[(2a + b)\left(a^2 + ab + 2b^2 + a(a + b)\right) - \left(a^2 + ab + 2b^2\right)(a + b)\right] \\
&> (a(a + b))^2.
\end{aligned}$$

Hence, $\frac{\partial J^{(1)}(y,a)}{\partial a} > 0$. The second term is

$$(y+b)\frac{\partial J(y,a)}{\partial a} = -2\frac{(y+b+a)\left(y(a+b)+a^2\right)}{(y+a)^2} < 0. \tag{E.13}$$

Therefore, $\frac{\partial G}{\partial a} > 0$. As the previous proof has established that $G(y,a)$ takes a U-shaped curve, $\frac{\partial G}{\partial a} > 0$ implies that its minimum point is monotonically decreasing in ω_s, concluding the first statement of point (i) in the lemma.

As $a \to 0$, it holds that $y_{min} \to 0$ and

$$\lim_{a\to 0} G(y_{min}, a) = \lim_{a\to 0}[2y_{min} - (y_{min} + b)] = -b < 0.$$

Also

$$\lim_{a\to\infty} G(y_{min}, a) = +\infty.$$

Therefore, together with

$$\frac{dG(y_{min}, a)}{da} = \frac{dy_{min}}{da}\frac{\partial G}{\partial y} + \frac{\partial G}{\partial a} > 0,$$

the above limits imply the existence of $\bar{a}$, such that, condition (E.11) holds if and only if $a < \bar{a}$. Given $\omega_{u,0}$, this result can be translated into ω_s so that $\tilde{\omega}_s = \frac{\omega_{u,0}}{\bar{a}}$. Noting that $a < \bar{a} \Leftrightarrow \omega_s > \tilde{\omega}_s$, we conclude the second half of point (i) in the lemma.

Suppose that condition (E.11) holds, meaning that y_0 and y_1 exist. For $j \in 0, 1$, $\frac{\partial G}{\partial a} > 0$ implies that $\frac{dy_0}{da} > 0$ and $\frac{dy_1}{da} < 0$. Also, the implicit function theorem leads to

$$\frac{dy_j}{da} = -\frac{1}{G^{(1)}(y_j, a)}\frac{\partial G(y_j, a)}{\partial a}.$$

By taking the total derivative of θ_k,

$$\begin{aligned}\frac{d\theta_k}{da} &= \frac{1}{2}\frac{1}{\sqrt{y_j}}\frac{dy_j}{da}G^{(1)}(y_j, a) + \sqrt{y_j}\frac{\partial J^{(1)}(y_j, a)}{\partial a} \\ &= -\frac{1}{2}\frac{1}{\sqrt{y_j}}\left(2y_j\frac{\partial J^{(1)}(y_j, a)}{\partial a} - \frac{\partial J(y_j, a)}{\partial a}\right) + \sqrt{y_j}\frac{\partial J^{(1)}(y_j, a)}{\partial a} \\ &= \frac{1}{2}\frac{1}{\sqrt{y_j}}\frac{\partial J(y_j, a)}{\partial a} \qquad\qquad (E.14) \\ &< 0,\end{aligned}$$

where the last inequality follows from (E.13). Therefore, we conclude that the boundary slopes, θ_H and θ_L are both increasing in ω_s.

Consider the magnitude of the reactions. By plugging (E.13) into (E.14),

$$\frac{d\theta_k}{da}| \propto \frac{\sqrt{y_j}}{(y_j + a)^2}\left(1 + \frac{a}{y_j + b}\right)\left(1 + \frac{1}{y_j}\frac{a^2}{a + b}\right).$$

It is straightforward that the last terms are all decreasing in y_j. Since we compare $y_0 < y_1$ with a being fixed, we obtain $\frac{d\theta_H}{d\omega_s} > \frac{d\theta_L}{d\omega_s}$, thereby concluding the proof.

Proof of Statements (iii) and (iv). Note that $\lim_{\omega_s \to \tilde{\omega}_s} y_j = y_{min}$, and $J(y_{min}) = K_{min}\beta_{min} = 2\beta_{min}J^{(1)}(y_{min})$ holds. Therefore, at this limit,

$$\theta'_H = \frac{J(\beta_{min}^2)}{2\beta_{min}} = \beta_{min}J^{(1)}(\beta_{min}^2) = \frac{K_{min}}{2} = \theta_k,$$

where the last equality holds because of (E.5). This concludes the proof of statement (iii).

Finally, consider the limit at $\omega_s \to 0$, which implies $a \to \infty$ (given $\omega_{u,0}$). Since $y_{min} \searrow a$, we can denote it as $y_{min} = a + \epsilon$, where $\epsilon > 0$ is a small positive and $\lim_{\omega_s \to \tilde{\omega}_s} \epsilon = 0$. With this notation, function J is rewritten at the limit as

$$\lim_{\omega_s \to \tilde{\omega}_s} J(y_{min}, a) = \lim_{a \to \infty, \epsilon \to 0} \frac{\epsilon(2a + b + \epsilon)}{(2a + \epsilon)(a + b + \epsilon)} = 0.$$

Since $\theta'_H = \frac{J(y_{min})}{2\sqrt{y_{min}}}$, we conclude that $\lim_{\omega_s \to \tilde{\omega}_s} \theta'_H = 0$.

As for statement (iv), case 1 occurs if, and only if, the curve $\theta = \frac{\varphi}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}$ exceeds θ_H at $\omega_s = \tilde{\omega}_s$. The above analyses have already established that $\theta_H = \frac{K^{(1)(y_{min})}}{2}$, where y_{min} is implicitly characterized by parameters a and b, we can define the threshold φ^* as

$$\varphi^* \equiv K^{(1)(y_{min})}\sqrt{\frac{\omega_{u,0}}{\bar{a}\omega_{u,1}}}. \tag{E.15}$$

□

Note that Proposition 4.1 follows from points (i) and (ii) of Lemma E.3, whereas Proposition **??** is shown by putting monotonically decreasing θ in equation (**??**) together with points (i) and (ii).

F Proof of Lemma ?? and Proposition 5.1

The optimal trading strategy satisfies the following condition.

$$\therefore (1 - 2\lambda_0\beta_0) + \frac{\psi}{2\lambda_1}\xi^2(\gamma^2\beta_0 + \gamma(1-\gamma)) = 0. \tag{F.1}$$

The marginal benefit of increasing s^* is given by

$$\mathrm{E}_\beta[v(\omega_s, \beta_0)] = \left[(1 - \lambda_0\beta_0) + \frac{\psi}{4\lambda_1}\xi^2(\gamma^2\beta_0 + 2\gamma(1-\gamma))\right]\beta_0 + \frac{\psi}{4\lambda_1}\xi^2(1-\gamma)^2 + (1-\psi)\frac{1}{2}\sqrt{\frac{\omega_{u,1}}{\omega_s}}. \tag{F.2}$$

Then, it holds that

$$\begin{aligned}\frac{\partial \mathrm{E}_\beta[v(\omega_s, \beta_0)]}{\partial\beta_0} &= \left[(1 - 2\lambda_0\beta_0) + \frac{\psi}{2\lambda_1}\xi^2(\gamma^2\beta_0 + \gamma(1-\gamma))\right] \\ &\quad + \left[-\beta_0\frac{\partial\lambda_0}{\partial\beta_0} + \frac{\psi}{4\lambda_1}\frac{\partial}{\partial\beta_0}\left[\xi^2(\gamma^2\beta_0 + 2\gamma(1-\gamma))\right]\right]\beta_0 + \frac{\psi}{4\lambda_1}\frac{\partial}{\partial\beta_0}\left(\xi^2(1-\gamma)^2\right).\end{aligned}$$

Note that the first term is zero due to the envelope theorem using (F.1). By using the notations introduced in equations (E.1), the remaining terms are summarized as follows.

$$-\beta_0 \frac{\partial}{\partial \beta_0} \frac{\beta_0}{\beta_0^2 + a} + \frac{\psi}{4\lambda_1} \left[\beta_0^2 \frac{\partial}{\partial \beta_0} \left(\frac{\beta_0}{a+b+\beta_0^2} \right)^2 + 2\beta_0 \frac{\partial}{\partial \beta} \left(\frac{1}{a+b+\beta_0^2} \right)^2 b\beta_0 + \frac{\partial}{\partial \beta} \left(\frac{b}{a+b+\beta_0^2} \right)^2 \right]. \tag{F.3}$$

From the proof in Appendix E, it holds that $y > a$ in the equilibrium. Therefore, the first term is negative. As for the second term, it holds that

$$\begin{aligned} \text{Second terms of (F.3)} &= 2\beta_0^3 \left(\frac{a+b-\beta_0^2}{(a+b+\beta_0^2)^3} \right) + 2\frac{a+b-3\beta_0^2}{(a+b+\beta_0^2)^3} b\beta_0 - 4b^2 \frac{\beta_0}{(a+b+\beta_0^2)^3} \\ &\sim -y^2 + y(a+b-3b) + b(a-b). \end{aligned}$$

Since $y > a$, the quadratic equation above is negative for all $y > a$ if and only if it is negative at $y = a$. This is true because

$$-a^2 + a(a-2b) + b(a-b) = -ab - b^2 < 0.$$

Therefore, the marginal benefit in (F.2) is decreasing in β_0, concluding the proof of Lemma **??**. Proposition 5.1 directly follows from Lemma **??** and equation (**??**).